nature chemical engineering

Article

https://doi.org/10.1038/s44286-026-00391-2

Redox-decoupled electrolysis for direct air capture of CO_2

Received: 1 May 2025

Accepted: 13 April 2026

Published online: 26 May 2026

Check for updates

Shijie Liu[1,7], Yurou Celine Xiao[1,7], Dongha Kim[1,2,7], Zunmin Guo[1,7], Eloi Grignon[3,7], Yuke Li[1], Ian Munroe[1], Hang Zhang[1], Jiexin Zhu[1], Zhizheng Wu[1], Jonathan P. Edwards[4], Jinqiang Zhang[5], Jieyuan Liu[1], Panagiotis Papangelakis[1], Yuxuan Che[1], Hyeon Seok Lee[1], Feng Li[1], Prasad V. Sarma[1], Qiyou Wang[1], Cai Wang[1], Todd Scheidt[1], Rui Kai Miao[1,8] ✉, Dwight Seferos[3,8], Yi Xu[6,8] ✉ & David Sinton[1,8] ✉

Electrochemical direct air capture (eDAC) leverages renewable electricity to remove atmospheric carbon dioxide (CO_2), offering an alternative to carbon-intensive thermal methods. However, existing eDAC systems achieve high energy efficiency only when producing a dilute hydroxide stream (pH ≈ 13) that is incompatible with current air contactors. Attempts to generate more concentrated capture solutions encounter the fundamental limitation of proton and hydroxide recombination, lowering the current efficiency and increasing energy requirements. Here we present a decoupled strategy whereby CO_2 liberation and sorbent regeneration are spatially separated, achieving high current and energy efficiency via redox-decoupled electrolysis. We tuned the redox mediator and synthesized a cation exchange membrane to ensure fast reaction kinetics, a low operating voltage and stability. The combined redox-decoupled approach achieved a capture-rate-normalized energy intensity of 0.22 GJ m^2 yr t^{-2} (at 50 mA cm^{-2}), a threefold improvement over previous work. Redox-decoupled eDAC provides an energy-efficient means of generating the concentrated alkaline capture solutions needed for large-scale direct air capture.

Direct air capture (DAC) is required to address legacy anthropogenic carbon dioxide (CO_2) emissions, including those from hard-to-abate sectors such as aviation and cement production[1–3]. Conventional DAC uses liquid sorbents based on aqueous hydroxide (OH^-) for their high capture rate, oxygen tolerance and recyclability[4,5]. However, the most developed OH^--based DAC method relies on a 900 °C thermal-swing process for CO_2 liberation and sorbent regeneration, and the substantial CO_2 emission associated with the thermal step reduces the overall effectiveness of DAC[6,7].

Replacing thermal regeneration with a renewably powered electrochemical process that operates at ambient temperature could lower emissions. Electrochemical direct air capture (eDAC) systems developed so far have achieved impressively low electrolyzer operating voltages and thereby high energy efficiencies[8–15]. However, these systems produce dilute capture solutions of insufficient OH^- concentration (<0.2 M, pH 13.3), leading to slow CO_2 capture kinetics that, in turn, require prohibitively large air contactors[8–10,12,14,16]. The air contactor is the dominant capital expenditure, and its size, which is determined

[1]Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Ontario, Canada. [2]School of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Republic of Korea. [3]Department of Chemistry, University of Toronto, Toronto, Ontario, Canada. [4]Deep Sky Corporation, Montreal, Quebec, Canada. [5]School of Mathematical and Physical Sciences, Faculty of Science, University of Technology Sydney, Sydney, New South Wales, Australia. [6]CAS Laboratory of Colloid and Interface and Thermodynamics, Institute of Chemistry, Chinese Academy of Sciences; School of Chemistry, University of Chinese Academy of Sciences, Beijing, China. [7]These authors contributed equally: Shijie Liu, Yurou Celine Xiao, Dongha Kim, Zunmin Guo, Eloi Grignon. [8]These authors jointly supervised this work: David Sinton, Rui Kai Miao, Yi Xu, Dwight Seferos. ✉e-mail: ruikai.miao@mail.utoronto.ca; yixusheldon@iccas.ac.cn; dave.sinton@utoronto.ca

by the CO_2 capture rate, directly impacts the economic viability of DAC[7,17]. Optimized industrial DAC operates with OH^- capture solutions that range from 1 to 2 M (pH 14–14.3) to achieve sufficient CO_2 capture rates. Indeed, the capture rate of a pH 14 solution is tenfold that of a pH 13.3 solution[18,19]. Small changes in pH thus substantially influence the capture kinetics and ultimately the cost of DAC.

A typical eDAC electrolyzer uses a pH-swing process to concentrate the capture solution in an alkaline compartment and release CO_2 in an acidic compartment (Fig. 1a). However, in this configuration, where the alkaline and acidic compartments are separated only by a single ion exchange membrane, it is challenging to achieve a high current efficiency (CE; the fraction of target ions transported across a membrane). At high concentrations, OH^- generated at the cathode competes with carbonate ions (CO_3^{2-}) for transport through the anion exchange membrane (AEM) into the acidic compartment (Fig. 1a left), and protons (H^+) compete with potassium ions (K^+) for transport through the cation exchange membrane (CEM) into the alkaline compartment (Fig. 1a right). These competing ion fluxes result in energy loss, as H^+ neutralizes with OH^- to form water rather than liberating CO_2. This effect becomes more pronounced at higher currents with high concentrations of OH^- and H^+ diffusing across the pH gradient[13,15,20].

We investigated this CE limitation by generating a concentrated OH^- capture solution (for example, 1 M potassium hydroxide (KOH)) using a three-compartment pH-swing configuration (Supplementary Fig. 1)[17,19]. With a 1-mm-thick middle compartment, we generated a 1 M KOH capture solution, but its CE was only 69% (Fig. 1b). We reasoned that the rapid H^+ diffusion through the middle compartment, occurring before H^+ could react to liberate CO_2, was the primary cause of this drop in CE. Increasing the middle compartment's thickness to 3 mm—thereby extending the H^+ diffusion length—improved the CE by 12%, although at the expense of a 50% higher electrolyzer voltage due to increased ohmic resistance (Supplementary Fig. 2). A recent approach using a polymer reinforcement layer within the ion exchange membrane similarly improved the CE, albeit at the cost of a higher electrolyzer voltage[15,21]. The high operating voltage leads to low energy efficiencies and high DAC operating costs.

Generating a sufficiently concentrated OH^- capture solution with high energy efficiency requires a wholly different electrolyzer design. We sought to overcome the fundamental challenge of the rapid H^+-diffusion-induced CE limitation by decoupling OH^- and H^+ generation into two distinct electrolyzers (P1 and P2; Fig. 1c and Supplementary Fig. 3), which provide sufficient time for H^+ to react with CO_3^{2-}. As a result, the P1 anode environment is weakly alkaline, which slows H^+ diffusion and enables a high CE with K^+ as the primary charge carrier (Supplementary Discussion 1 and Supplementary Figs. 4–9). To achieve rapid reaction kinetics, a high CE and a low operating voltage, we used redox mediator (RM) molecules, which can facilitate ionic charge transfer between the decoupled electrolyzers, and synthesized a selective CEM. We further combined density functional theory (DFT) and experimental results to optimize the RM, that is, sulfate-substituted 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO-S), to achieve a highly energy efficient and stable eDAC operation.

Redox-decoupled pH-swing mechanism

Although the two electrolyzers P1 and P2 are spatially decoupled, RMs can be used as charge carriers between the anode of P1 and the cathode of P2. An ideal RM for this application would be stable, exhibit fast reaction kinetics and have a redox potential below 1.23 V versus the standard hydrogen electrode (SHE) to avoid undesirable oxygen generation at the P1 anode. This sets a practical lower limit for P1 full-cell voltage at −2 V (Supplementary Discussion 2 and Supplementary Fig. 10). Drawing inspiration from aqueous redox flow batteries—an established and scalable system—we used a ferricyanide/ferrocyanide ($[Fe(CN)_6]^{3-}$/$[Fe(CN)_6]^{4-}$) couple in the redox-decoupled configuration.

During operation, CO_3^{2-} was added to the CO_2 liberation solution, and CO_2 was released via a pH swing driven by H^+ generated in P2. We delineated the pH limits of the CO_2 liberation loop and the P1 anode operating window required for a high CE. When the liberation solution was pH > 2, H^+ transport was suppressed (Supplementary Discussion 1 and Supplementary Figs. 4 and 5). Owing to OH^- crossover and CO_3^{2-} buffering, the local pH of the P1 anode was maintained above 10 (Supplementary Figs. 6–9). The weakly alkaline environment promoted K^+ as the primary charge carrier and resulted in a 1 M KOH capture solution with an average CE of 91% (Fig. 1d and Supplementary Figs. 11 and 12). The redox-decoupled configuration increased the CE markedly, generating a highly concentrated OH^- capture solution and saving over 33% in the overall system energy compared with prior approaches producing a similar concentrated stream (Supplementary Fig. 13).

However, at high current densities, we observed a rapid increase in the P2 voltage, resulting in a high full-cell voltage (that is, the sum of the P1 and P2 voltages) and a high overall system energy (Supplementary Fig. 14). We attribute this to the ferricyanide/ferrocyanide couple's overall charge of −3/−4, which experiences electrostatic repulsion at the negatively charged reduction electrode in P2 (Supplementary Fig. 15). Increasing the RM concentration slightly improved the P2 voltage, but the full-cell voltage still increased steeply at higher current densities (Supplementary Fig. 14). We instead shifted our focus to organic RMs, whose design flexibility enables the incorporation of substituents to tailor the net charge and overcome electrostatic barriers during both reduction and oxidation processes[22–24]. These substituents can also influence the overall electrochemical performance by tuning the RM's aqueous solubility, redox potential, kinetics and stability[25,26].

We screened RM candidates with low binding energies to the electrically conductive electrode, which avoids deposition and minimizes electrolyzer overpotentials[12,27–29]. We evaluated four common classes of organic RM—that is, 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO), phenazine, quinone and viologen—via DFT simulations (Fig. 2a, Supplementary Discussion 3 and Supplementary Figs. 16–20). The simulation results showed that phenazine, quinone and viologen rely on their large aromatic structures to stabilize their redox-active states, leading to strong inter- and intramolecular π–π stacking interactions with the carbon electrodes[30–32]. By contrast, the sp^3 carbon core of TEMPO results in the lowest binding energy to the electrode due to weak London dispersion forces[30,33], preventing deposition (Supplementary Fig. 21). TEMPO is therefore a suitable organic RM candidate as it exhibits a reversible one-electron redox mechanism with a low redox potential (0.75 V versus SHE), stabilized by the steric hindrance that arises from the four methyl groups adjacent to the nitroxyl group (Fig. 2b)[34–36]. Compared with the other organic RMs, TEMPO is compact, has low toxicity, is relatively inexpensive and can be produced at scale[37–39].

The redox potential, reaction kinetics and molecular stability of TEMPO derivatives are influenced by steric hindrance, charge distribution, net molecular charge and ion pairing[26,40–44]. We performed DFT calculations on five TEMPO derivatives with various substituents to identify candidates that support energy-efficient redox reactions[35,36,45,46]. Whereas the substituents have a negligible effect on the steric hindrance at the TEMPO core, derivatives with smaller substituents exhibit improved mass transport and enhanced reaction kinetics (Supplementary Discussion 3 and Supplementary Fig. 22)[40]. To achieve fast reaction rates, high mobility and excellent water solubility, we assessed three TEMPO monomers with small polar substituents: hydroxyl-substituted (TEMPO-OH), sulfate-substituted (TEMPO-S) and phosphate-substituted (TEMPO-P) (Fig. 2c and Supplementary Figs. 23–30)[47–49]. All three TEMPO derivatives demonstrated chemical compatibility within the pH range of 2–12 (Supplementary Figs. 31–38).

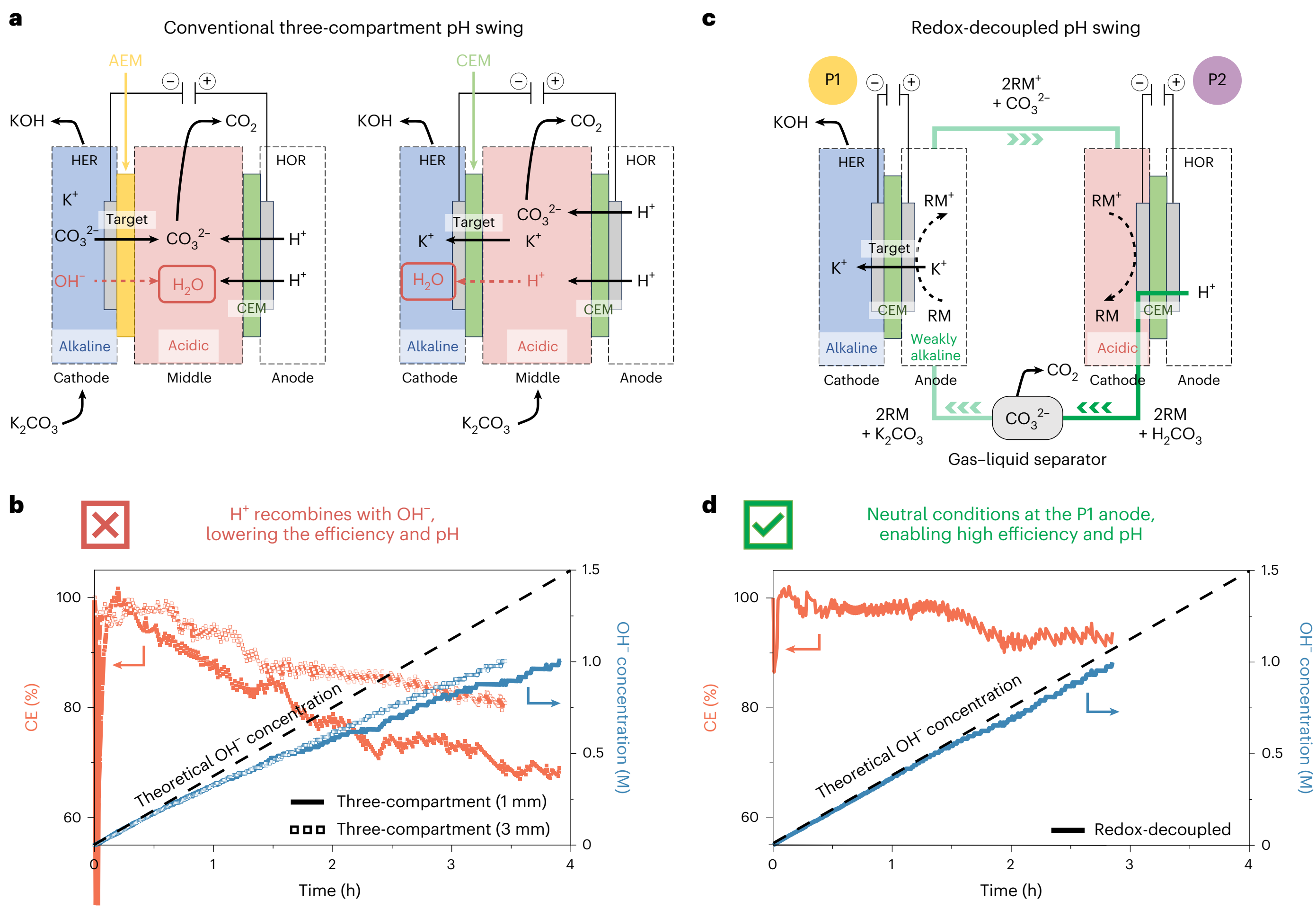

Fig. 1 | Conventional three-compartment pH swing versus redox-decoupled pH swing. a, Schematic illustrating CE losses caused by H^+ and OH^- recombination in a conventional three-compartment pH-swing configuration. In the AEM + CEM configuration (left), the local pH at the cathode catalyst becomes highly alkaline, generating a high concentration of OH^-. The OH^- electromigrates through the AEM as the charge carrier and neutralizes electrochemically produced H^+, leading to CE losses. In the CEM + CEM configuration (right), because H^+ is smaller and more mobile than alkali metal cations, H^+ migrates more readily through the CEM. This enables direct H^+ crossover from the anode compartment to the cathode compartment, where it neutralizes electrochemically generated OH^-, leading to CE losses. HER, hydrogen evolution reaction; HOR, hydrogen oxidation reaction. **b**, The CE and OH^- concentration of the KOH capture solution produced in a conventional three-compartment pH-swing configuration using a middle compartment with a thickness of 1 mm or 3 mm. **c**, Schematic illustrating that high CE and alkaline pH are achieved by separating the alkaline and acidic compartments into two distinct electrolyzers, thereby spatially decoupling OH^- and H^+ generation in a redox-decoupled pH-swing configuration. H_2CO_3, carbonic acid. **d**, The CE and OH^- concentration of the KOH capture solution produced using the redox-decoupled pH-swing configuration.

In an H-cell, we then performed cyclic voltammetry (CV) to investigate their charge and discharge behaviors (Supplementary Figs. 39–41). TEMPO-S showed the narrowest peak-to-peak separation (ΔE_p), indicating the fastest charge-transfer kinetics (Supplementary Tables 1–3). Its high oxidation and reduction currents (I_{oxid} and I_{red}, respectively) indicated faster mass-transfer rates compared with TEMPO-OH and TEMPO-P (Supplementary Figs. 42 and 43 and Supplementary Tables 4–6). TEMPO-S and TEMPO-P exhibited a wider electrochemical reversible pH range (2–12) (Supplementary Figs. 44–46). TEMPO-S also showed faster kinetics and improved stability relative to TEMPO-OH (Supplementary Figs. 47 and 48)[50,51]. We attribute this result to the negatively charged sulfate ($-OSO_3$) and phosphate ($-OPO_3$) substituents that electrostatically repel OH^- ions.

Considering charge distribution, ion pairing and net charge, CV and DFT showed that TEMPO-S lowers the ring electron density, accelerates electron transfer and suppresses undesirable side reactions, thereby broadening the operational window (Supplementary Discussion 4 and Supplementary Figs. 49–54). TEMPO-S also carries a favorable net charge of −1/0, promoting efficient oxidation at P1 and mass transport under optimized flow. The reaction kinetics are a critical parameter that governs the energy efficiency of the redox processes. A comparison of the diffusion coefficients of TEMPO (D_{TEMPO}) and $TEMPO^+$ (D_{TEMPO^+}) and the electron-transfer rate constant (k^0) supports TEMPO-S as the ideal RM candidate (Fig. 2d, Supplementary Discussion 5 and Supplementary Table 7). We therefore selected TEMPO-S for the redox-decoupled system.

Using electron paramagnetic resonance (EPR) spectroscopy, we determined that an optimal TEMPO-S concentration of 0.1 M avoids collision-induced structural decomposition (Fig. 2e)[52,53]. To maintain efficient and stable TEMPO-S redox reactions, we optimized the solution flow rate, composition and pH. A parametric sweep of the capture and CO_2 liberation flow rates identified 50 ml min^{-1} (liberation loop) and 25 ml min^{-1} (capture loop) as the operating points that minimize the full-cell voltage with stable operation (Supplementary Figs. 53 and 55). A 50:50 ratio of TEMPO-S to $TEMPO\text{-}S^+$ produced the lowest full-cell voltage—better than the best ratios for TEMPO-OH and TEMPO-P (Fig. 2f and Supplementary Figs. 56–59).

Prior reports indicate that the stability of TEMPO can be sensitive to extreme pH (Supplementary Fig. 60)[50,51,54–56]. Accordingly, we maintained the TEMPO-S-containing electrolyte within a moderate

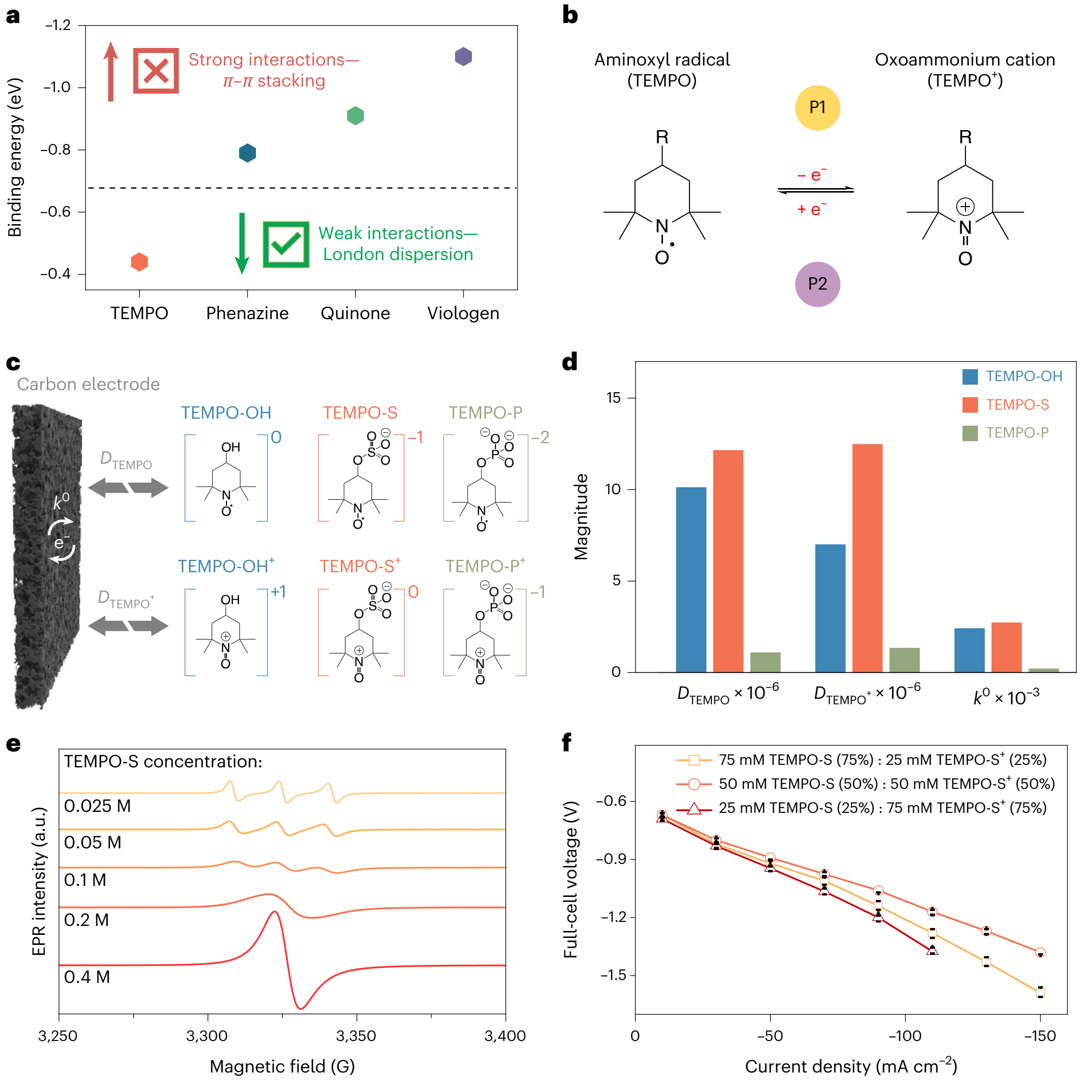

Fig. 2 | Rational selection of TEMPO-S for fast mass transfer and charge-transfer kinetics. a, DFT-calculated binding energy of TEMPO, phenazine, quinone and viologen molecules on a carbon electrode. **b**, Redox mechanism of TEMPO, which undergoes oxidation (P1) and reduction (P2) to reversibly store and release electrons (e^-). **c**, Schematic illustration of the electrochemical redox reactions of three TEMPO derivatives. **d**, Summary of D_{TEMPO}, D_{TEMPO^+} and k^0 values for the three TEMPO derivatives. **e**, EPR spectra of TEMPO-S at concentrations ranging from 0.025 M to 0.4 M. **f**, Full-cell voltage of the redox-decoupled system, using different TEMPO-S:TEMPO-S$^+$ ratios. The data points are presented as the mean value ± 0.03 (n = 3).

bulk pH window (3–11) during operation to balance the kinetics and mediator stability. During operation, the electrolyte remained near neutral (Supplementary Fig. 5). TEMPO-S is electrochemically stable within a pH range of 2–12 (Supplementary Figs. 47 and 48). The selected operating window therefore provides additional tolerance to local pH gradients near the electrode surface, helping to suppress TEMPO-S degradation. A CO_3^{2-} concentration of 0.05 M regulated the pH of the liberation solution, thereby prolonging the stability of TEMPO-S and optimizing the performance of the redox-decoupled system (Supplementary Fig. 61)[54–56].

However, when testing the system with a conventional CEM (that is, Nafion), we observed severe TEMPO-S crossover into the cathode compartment (Fig. 3a and Supplementary Figs. 62b and 63a)[57–59]. We hypothesized that reducing the channel size of the membrane would mitigate TEMPO-S crossover due to its transport via a vehicle mechanism, and that increasing the degree of negatively charged sulfonate groups would promote K^+ mass transfer due to its transport via a hopping mechanism (Supplementary Fig. 64)[60–62]. Casted sulfonated poly(ether ether ketone) (SPEEK), with narrow channels and twice as many sulfonate sites as Nafion, effectively reduced TEMPO-S crossover (Fig. 3b and Supplementary Figs. 62c and 63b). However, its lower cation conductivity increases the membrane ohmic drop (Supplementary Fig. 65)[63–66]. Using a 15-µm-thick SPEEK membrane balances crossover and resistance, maintaining a TEMPO-S crossover reduction of ~25-fold while lowering the full-cell voltage (Fig. 3c,d, Supplementary Discussion 6 and Supplementary Figs. 66–68).

DAC demonstration via the redox-decoupled system

The redox-decoupled system for energy-efficient eDAC optimized the RM, the composition of the CO_2 liberation solution and the CEM used. The system resulted in a minimum energy consumption of 0.67 GJ tCO_2^{-1}, based on the potential difference in an H-cell CV scan (Supplementary Discussion 5 and Supplementary Fig. 69). The system maintained a CE of ~90% over a current density operating range of 10–200 mA cm^{-2}, with an energy consumption of 4.0–8.7 GJ tCO_2^{-1} and a full-cell voltage of 0.86–1.71 V to generate a 1 M KOH capture solution (Supplementary Discussion 7 and Supplementary Figs. 70 and 71). However, above 150 mA cm^{-2}, the full-cell voltage increases, which we attribute primarily to elevated interfacial resistance that arises from

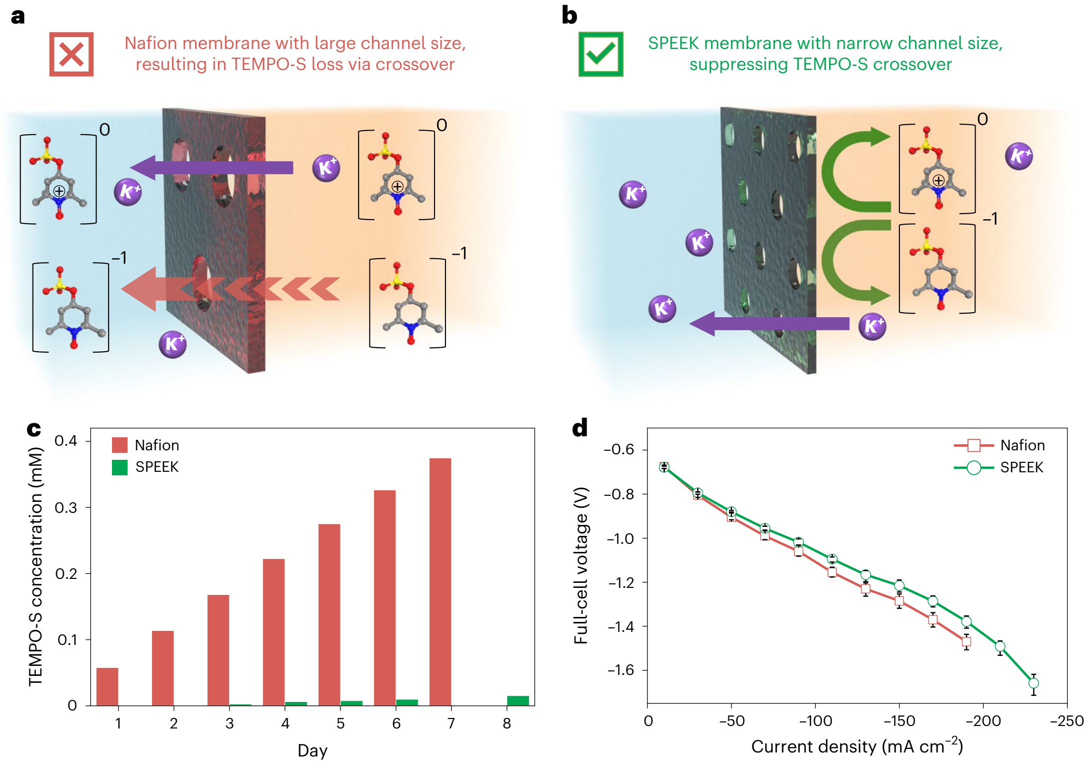

Fig. 3 | CEM with a narrow membrane channel size suppresses TEMPO-S crossover. a, Schematic illustrating TEMPO-S crossover through a Nafion membrane, which has a relatively large channel size that allows mediator crossover. Atom color code: C, gray; N, blue; O, red; S, yellow; H atoms are omitted for clarity. **b**, Schematic illustration of a casted SPEEK membrane with a narrow channel size that suppresses TEMPO-S crossover. **c**, TEMPO-S concentration permeating a conventional Nafion membrane or a cast SPEEK membrane over one week, quantified using UV-vis spectrophotometry. **d**, Full-cell voltage of the redox-decoupled system, using either a conventional Nafion membrane or a cast SPEEK membrane. The data points are presented as the mean value ± 0.05 ($n = 3$).

the formation of CO_2 bubbles. We detected negligible oxygen evolution within the current density operating window, demonstrating TEMPO's fast kinetics and the feasibility of the redox-decoupled system (Supplementary Fig. 72)[36,67].

To demonstrate capture–release cycling fully, we integrated the redox-decoupled system with a crystallizing air capture unit that precipitated solid potassium carbonate (K_2CO_3) directly from atmospheric CO_2 with a precipitate efficiency ($\eta_{precipitate}$) of 93.4 ± 1.2% (Supplementary Discussions 8–11, Supplementary Figs. 73 and 74 and Supplementary Table 8)[68]. The redox-decoupled system operated for over 200 hours, producing highly concentrated OH^- capture solutions for ten cycles at 50 mA cm^{-2} (Fig. 4a and Supplementary Figs. 75–77). This average full-cell voltage was 1.08 ± 0.15 V, corresponding to an average specific energy consumption of 5.4 ± 0.7 GJ tCO_2^{-1} (Fig. 4b).

We observed a net water transfer of 1.2 ± 0.1 ml per 20 h cycle due to the electro-osmotic drag of K^+. During each cycle, although the increase in capture solution pH led to a reduction in the flow rate of CO_2 released and a longer time required for complete CO_2 liberation, the average CE for producing 1 M KOH remained at 92 ± 1% throughout ten cycles (Fig. 4c–e). We measured negligible TEMPO-S crossover, with an average apparent decay rate of 0.046% h^{-1} (Supplementary Figs. 78–80). Mass spectrometry detected hydroxylamine (N–H) peaks, which were attributed to the H^+ flux in P2 cleaving the aminoxyl radicals (N–O·) to N–H (Supplementary Fig. 81)[36,39]. However, the N–O· functionality is chemically recoverable from N–H using an oxidizing agent (for example, hydrogen peroxide)[69,70]. The redox-decoupled configuration uses commercially available platinum catalysts and carbon electrodes, both of which have proved to be robust and durable under extreme conditions[71]. X-ray photoelectron spectroscopy analysis revealed no apparent changes in surface composition or electrode morphology after 200 hours of electrolysis (Supplementary Fig. 82). The combination of high OH^- concentration and improved efficiency resulted in a capture-rate-normalized energy intensity (that is, the system energy normalized by the CO_2 capture rate) of 0.2–0.4 GJ m^2 yr t^{-2} over the current density range of 10–200 mA cm^{-2} (Supplementary Table 10).

Comparison with existing approaches, and future improvements

The redox-decoupled eDAC system therefore offers a practical and efficient route for generating KOH capture solutions of over 1 M and with a high CE. The system can be powered by renewable electricity, and combines continuous operation with established and robust components, including commercial electrodes and a high-yield synthesized RM, which facilitates scale-up and simplified operation[6,7]. Recent advances in three-compartment eDAC systems have reduced the full-cell voltage by thinning the middle compartment thickness or adopting porous solid electrolytes[15,72], but the close proximity of the acidic and alkaline compartments limits their CE at high capture-solution concentrations and poses scale-up challenges in reliably sealing thin separation challenges. Compared with the alternating electrocatalysis configuration, the present system operates continuously, doubling the production rates of OH^- and H^+, eliminating the need for expensive iridium, and leveraging a simple, high-yield synthesis for TEMPO-S[9,12]. Whereas both aqueous and nonaqueous H^+-coupled electron-transfer systems offer lower theoretical energies, they are commonly limited by low operating current densities[8,73].

Developing oxygen-tolerant RMs also remains a key requirement for robust, scalable eDAC. A techno-economic analysis of the redox-decoupled configuration estimated a net cost of US$289 t$CO_2^{-1}$, with

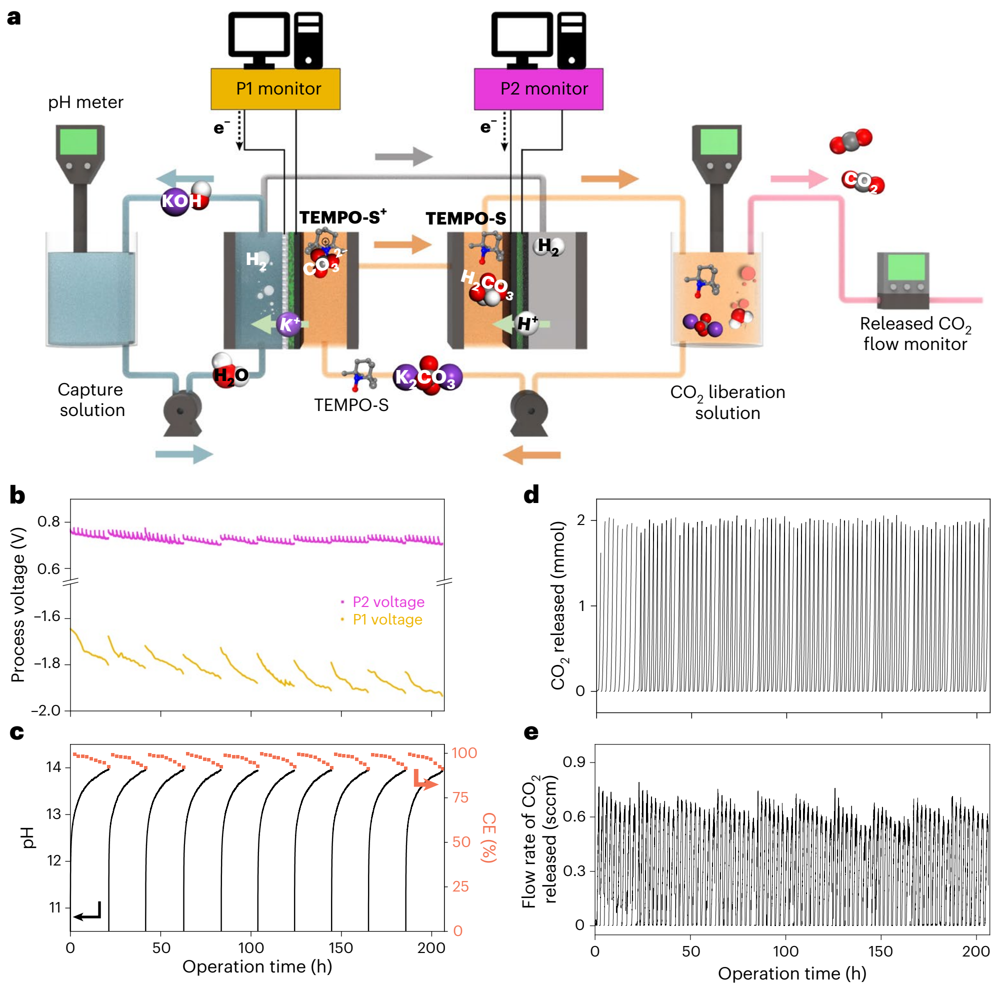

Fig. 4 | eDAC performance of the redox-decoupled system over 200 hours. **a**, Schematic illustration of the system configuration and experimental set-up for redox-decoupled eDAC operation. **b**, Voltage profiles of the P1 and P2 processes during operation at 50 mA cm^{-2}. **c**, pH of the regenerated capture solution (left axis) and the corresponding system CE (right axis) during operation. **d**, Total amount of CO_2 released during operation. **e**, Flow rate of CO_2 released during operation.

operating expenditures accounting for 62% of the total cost (Supplementary Discussion 12). The redox-decoupled system would benefit from further improvements in mediator stability and kinetics, as well as operation at a high current density to reduce the electrolyzer area. Advances in gas management and interfacial engineering are expected to further support low-voltage operation at high current densities (Supplementary Fig. 71 and Supplementary Table 11)[36,67].

In a future pilot-scale system, the CO_2 liberation solution tank can be integrated with automated mixing and metered K_2CO_3 dosing. Valves, level sensors and closed-loop control can regulate the electrolyte volume and pH and maintain the RM concentration. These standards are well established balance-of-plant components in large-scale redox flow battery systems that improve operational consistency and may enhance mediator lifetime, while adding modest cost relative to the electrochemical stack[74,75]. We acknowledge that large-scale implementation will inevitably require the careful management of chemical feedstocks, mediator stability and operational logistics. These aspects merit further study and will be important considerations in future system design.

Conclusion

In summary, the redox-decoupled approach separates the acidic CO_2 liberation step from the alkaline sorbent regeneration, generating industrially compatible KOH capture solutions with high CE. By combining the robustness of inorganic OH^--based CO_2 capture with redox electrochemistry's favorable kinetics, the redox-decoupled eDAC approach surpasses prior eDAC strategies (Supplementary Fig. 84 and Supplementary Table 10). Redox-decoupling thus presents a means to overcome the fundamental CE limitation in generating concentrated capture solutions that are compatible with industrial DAC, providing a pathway to the widespread application of atmospheric CO_2 capture.

Methods

Preparation of catalysts and chemicals

All chemicals used for synthesis of the organic redox-active compounds and preparation of the electrolytes were purchased from Sigma Aldrich, and included 4-hydroxy-TEMPO (TEMPO-OH; 95%), potassium hexacyanoferrate(II) trihydrate (ACS reagent, >98.5%), potassium hexacyanoferrate(III) (ACS reagent, >99%), potassium sulfate (K_2SO_4; ACS reagent, >99%), K_2CO_3 (ACS reagent, >99%), KOH (ACS reagent, >85%), ethanol (ACS reagent, >99.5%), sulfuric acid (H_2SO_4; ACS reagent, 95–98%), ethyl acetate (ACS reagent, >99.5%), acetone (ACS reagent, >99.5%), phosphorus(V) oxychloride ($POCl_3$; 99%), diethyl ether (ACS reagent, >99%), *N*,*N*-dimethylformamide (ACS reagent, >99.8%) and isopropanol (99.5%). All aqueous electrolytes and solutions were prepared using Milli-Q-grade deionized (DI) water (18.2 MΩ). All of the chemicals used for electrode preparation were purchased from Fuel Cell Store, and included hydrophilic carbon paper (AvCarb MGL190), hydrophobic carbon paper (Freudenberg H23C3), PtC catalyst (60% platinum on Vulcan XC-72R), commercial CEM (Nafion 115, thickness 127 µm) and ionomer binder (Nafion 1,100 W). All electrocatalytic electrodes were prepared by spray-depositing an ink with a formulation PtC catalyst (1 mg ml^{-1}) and ionomer binder materials (5 wt%) in isopropanol (60 wt%). The target mass loading of the platinum catalyst was tuned to 0.5 mg cm^{-2}. The electrodes were dried for 24 h prior to experimentation. The area of the H-cell platinum mesh counter electrode was 4 cm^2. All of the chemicals used for the cast CEM were purchased from eSpin Technologies, including the SPEEK #4 polymer. The polypropylene line for the customized crystallizing capture unit was purchased from Amazon.

Synthesis of TEMPO-S

TEMPO-OH (4-hydroxy-2,2,6,6-tetramethylpiperidin-1-oxyl; 5 g, 29 mmol) was ground in a mortar to obtain a fine powder. This powder was then added in small increments to concentrated H_2SO_4 (35 ml) in a round-bottom flask with stirring. The TEMPO-OH powder dissolved slowly to form a viscous orange solution. After 30 min, the solution was transferred to an addition funnel and added dropwise to sodium carbonate (75 g) in water (550 ml) at 0 °C for neutralization. After stirring for 30 min at room temperature, the mixture was extracted twice with ethyl acetate to remove any organic-soluble impurities. The aqueous phase was then evaporated under vacuum to afford a light-pink solid. Acetone was added and the mixture was stirred for 1 h to extract TEMPO-S (1-oxyl-2,2,6,6-tetramethylpiperidin-4-yl sulfate) from the solid. The solid phase was then removed through vacuum filtration and washed with additional acetone to extract any residual product. The acetone was evaporated under vacuum to yield pure TEMPO-S as a dark-red solid (7.50 g, 94%)[76].

^{1}H NMR (400 MHz, DMSO-d6, quenched with phenylhydrazine): δ 4.32–4.41 (m, 1H), 1.94–1.99 (m, 2H), 1.35 (t, J = 11.9 Hz, 2H), 1.06 (d, J = 9.0 Hz, 12H). ^{13}C NMR (125 MHz, DMSO-d6, quenched with phenylhydrazine): δ 68.82, 58.58, 45.76, 33.01, 20.66.

Synthesis of TEMPO-P

$POCl_3$ (4.86 ml, 52.2 mmol) was dissolved in diethyl ether (100 ml) in a round-bottom flask. A solution of TEMPO-OH (6 g, 34.8 mmol) and triethylamine (4.87 ml, 34.8 mmol) in diethyl ether (90 ml) was transferred to an addition funnel and added dropwise to the mixture at 0 °C, giving off white fumes and forming a precipitate. After 16 h, the precipitates were filtered off and water (50 ml) was added to the mixture, which was stirred for another 10 min. The biphasic mixture was transferred to a separating funnel and the aqueous phase was drained off. The organic phase was extracted twice more with water and then discarded. The combined aqueous phases were then neutralized with 6 M sodium hydroxide, and the water was removed under vacuum. Ethanol was added to the solid mixture to redissolve the product, followed by filtration through a Celite plug. Removal of the solvent under vacuum afforded TEMPO-P (1-oxyl-2,2,6,6-tetramethylpiperidin-4-yl phosphate) as a red solid (4.34 g, 42%)[76].

^{1}H NMR (400 MHz, DMSO-d6, quenched with phenylhydrazine): δ 4.34–4.19 (m, 1H), 2.21–1.81 (m, 2H), 1.33 (t, J = 11.8 Hz, 2H), 1.07 (d, J = 4.5 Hz, 12H). ^{13}C NMR (125 MHz, DMSO-d6, quenched with phenylhydrazine): δ 66.97, 61.82, 47.15, 32.48, 21.94. ^{31}P NMR (162 MHz, DMSO-d6, quenched with phenylhydrazine): δ 0.51.

Chemical and electrochemical stability of TEMPO derivatives

To assess the chemical stability of the TEMPO derivatives across the pH range of 2–12, ^{1}H NMR spectra, ultraviolet-visible spectrophotometry and high-resolution mass spectrometry were used (Supplementary Figs. 31–38). For all samples, the pH level was adjusted using KOH and H_2SO_4.

To assess the electrochemical reversibility of the TEMPO derivatives across the pH range of 2–12, CV measurements were carried out using an H-type electrolytic cell (Supplementary Figs. 44–46). The working electrode was hydrophilic carbon paper (AvCarb MGL190). The working electrolyte was protected under N_2 during the measurements. A 50:50% TEMPO:$TEMPO^+$ ratio was tuned via galvanostatic electrocatalytic operation before the CV measurements. All profiles were recorded at the scan rate of 100 mV s^{-1} for 150 cycles.

To assess the electrochemical stability of the TEMPO-OH and TEMPO-S derivatives at pH 2 and pH 12, a galvanostatic charge–discharge flow battery configuration was used (Supplementary Figs. 47 and 48). The working electrode was hydrophilic carbon paper (AvCarb MGL190). The working electrolyte was protected under N_2 during the measurements. The volume of the working electrolyte used (0.1 M TEMPO) was 5 ml. The volume of the counter electrolyte used (0.2 M ferricyanide/0.2 M ferrocyanide in 0.25 M K_2SO_4) was 20 ml.

Cast SPEEK CEM

The customized SPEEK membranes were prepared via a solution-casting method. The dry SPEEK polymer (32 mg) was added in *N*,*N*-dimethylacetamide (5 ml) until the mixture was homogeneous. The mixture was poured into a 70-mm-diameter glass Petri dish and placed in a vacuum oven at 80 °C for 20 h, after which the temperature was increased to 120 °C for 6 h under vacuum conditions of −0.9 bar. The as-prepared dry membrane (~15-µm-thick) was soaked in 1 M H_2SO_4 for 24 h and in DI water for 1 h before use.

Electrocatalysis H-cell configuration

A 15 ml H-type electrolytic cell (Tianjin Aida Hengsheng Technology Development Co.) was used for the experiments. A glassy carbon electrode (3-mm-diameter) served as the working electrode, unless otherwise specified. A 3.5 M KCl Ag/AgCl electrode was used as the reference electrode, and a platinum mesh electrode functioned as the counter electrode. The working and counter compartments were separated using a Nafion 115 membrane. The counter electrolyte contained a 0.5 M $KHCO_3$ aqueous solution, unless otherwise specified. An N_2 flow rate of 10 sccm (standard cubic centimeters per minute) was purged during the operation.

To obtain different TEMPO:$TEMPO^+$ ratios, a hydrophilic carbon electrode with an area of 4 cm^2 was used to perform the electrochemical oxidation reaction to convert TEMPO into $TEMPO^+$. The total concentration of $TEMPO^+$ generated was determined using Faraday's law.

Hydroxide crossover CEM test configuration

The electrolyzer configuration comprised a custom-made electrolyzer cell. The electrocatalytic active area was 1 × 1 cm. A PtC catalyst on a hydrophilic carbon electrode was installed in the cathode compartment. A Nafion 115 CEM separated the P1 cathode and anode compartments. A bare hydrophilic carbon electrode was placed in the anode compartment. Various rubber gaskets were used to prevent liquid leakage. The catholyte was 1 M KOH (10 ml). The anolyte was 0.1 M $K_3Fe(CN)_6$ and 0.1 M $K_4Fe(CN)_6$ in 0.25 M K_2SO_4 aqueous solution (10 ml). The catholyte flow rate was 25 ml min^{-1}, and the anolyte flow rate was 50 ml min^{-1}. A pH meter was placed at the anolyte to monitor the increase in OH^- concentration over time.

Galvanostatic charge–discharge flow battery configuration

The flow battery configuration used a custom-made electrolyzer cell. The electrocatalytic active area was 1 × 1 cm. A bare hydrophilic carbon electrode was placed in the working compartment. A bare hydrophilic carbon electrode was installed in the counter compartment. A cation membrane separated the working and counter compartments. Various rubber gaskets were used to prevent liquid leakage. The flow rate of the working compartment was 60 ml min^{-1} to eliminate mass-transfer limitations.

Redox-decoupled configuration

The redox-decoupled configuration comprised two sets of custom-made electrolyzer cells (Supplementary Fig. 3). The electrocatalytic active area was 1 × 1 cm. For P1, PtC on a hydrophilic carbon electrode was installed in the cathode compartment. A CEM (for example, Nafion 115 or cast SPEEK) separated the P1 cathode and anode compartments. A bare hydrophilic carbon electrode was placed in the anode compartment. For P2, a bare hydrophilic carbon electrode was placed in the cathode compartment. A Nafion 115 membrane separated the P2 cathode and anode compartments. PtC on a hydrophobic carbon electrode was installed in the anode compartment. Various rubber gaskets were used to prevent liquid leakage. The capture solution and CO_2 liberation solution were applied through separate silicone tubes that were each connected to a peristaltic pump. The flow rate of the capture solution was 25 ml min^{-1}, and the flow rate of the CO_2 liberation solution was 50 ml min^{-1}. Both solutions were pumped into the bottom of each chamber and exited from the top and flowed back to their bulk electrolyte tank to form a closed cycle. Before starting the experiment, the gas–liquid separator was sealed and protected with N_2 gas.

Electrochemical measurements

All electrochemical tests were carried out at room temperature (~298 K) and atmospheric pressure (~1 atm). All electrochemical experiments were operated and recorded using electrochemical workstations (Autolab PGSTAT302N).

The CV measurements were carried out using an H-type electrolytic cell with the electrochemical workstation. The working electrolyte was protected under N_2 before and during the measurements. The different ratios of TEMPO:TEMPO$^+$ were tuned via the galvanostatic electrocatalytic operation. All profiles were recorded at different scan rates ranging from 10 to 200 mV s^{-1}. Allelectrode potentials (E) were measured using the Ag/AgCl reference electrode and were converted to the SHE scale using equation (1):

$$E(\mathrm{SHE}) = E(\mathrm{Ag/AgCl}) + 0.205. \quad (1)$$

Linear scan voltammetry measurements were conducted in an electrolytic cell using the electrochemical workstation. The cathode catalyst used PtC on hydrophilic carbon paper, and the anode catalyst used bare hydrophilic carbon paper. The profile was recorded at a scan rate of 10 mV s^{-1}.

Electrochemical impedance measurements were conducted in an electrolytic cell using the electrochemical workstation. The cathode catalyst used PtC on hydrophobic carbon paper, and the anode catalyst used PtC on hydrophobic carbon paper. The Nafion 115 and the cast SPEEK membrane were used as the CEM. Data points were acquired in the frequency range from 10^5 Hz at 100 mA cm^{-2} to 0.1 Hz at 100 mA cm^{-2}.

During operation of the redox-decoupled system, two independent electrochemical workstations were used to control and monitor P1 and P2. Electrochemical data were recorded at 10-s intervals, except for the stability operation, where the data were recorded 30-s intervals. All voltages reported are full-cell voltages without IR (I, current; R, resistance) compensation. Calculation of the current densities uses the geometric surface area of the electrode.

During the galvanostatic charge–discharge TEMPO-S flow battery operation, one electrochemical workstation was used to control and monitor the electrochemical performance of the flow battery. The working electrolyte was protected under N_2 during the measurements. To establish a steady state in the flow battery operation, five full cycles were performed with a charge cut-off voltage of 1 V and discharge cut-off voltage of −0.5 V. The state of charge (SOC) was estimated on the basis of the discharge operation time. To evaluate the oxygen evolution reaction at current densities ranging from 50 to 200 mA cm^{-2}, the charge operation time was set to 50% SOC at the current density, and the discharge operation was set with a discharge cut-off voltage of −0.5 V. The coulombic efficiency was calculated on the basis of the ratio between the discharge operation time and the charge operation time.

Experimental details of 200-h stability testing

During the 200-hour stability operation of the redox-decoupled system, the capture solution tank was filled with DI water (40 ml) prior to the operation. The gas–liquid separator was equipped with a 40-ml volume of the CO_2 liberation solution (that is, 0.1 M TEMPO-S with 0.25 M K_2SO_4 and 0.05 M K_2CO_3). Both the P1 and P2 voltage data were recorded, using the electrochemical workstations, every 10 s with P1 operating at −1.80 ± 0.14 V and P2 operating at 0.72 ± 0.05 V. pH meters (Hanna HI2020 edge) were used to simultaneously to monitor the solution pH every 60 s, and the actual pH values were converted using a pH calibration curve at various concentrations. The flow rates of the released gaseous CO_2 were recorded every 60 s using a flowmeter (Omega FMA-LP1615A) coupled with a customized vacuum pump (CO2Meter CM-0111) (Supplementary Fig. 77). The flow-meter accurately detects flow rates in the range of 0–5 sccm with an accuracy of ±0.8%. The recorded data were transmitted via a data logger (Omega OM-CPVOLT101A and OM-CP-IFC200). N_2 was used to protect the TEMPO solution during pause periods, while the N_2 supply was halted during electrolysis to ensure that all recorded data represented only the released CO_2.

Ten cycles of electrochemical operation were conducted to produce a highly concentrated OH^- capture solution. At the end of each round, the electrochemical cell, pH recorder and CO_2 flow-meter were paused, while the capture solution continued to recirculate. To maintain a 0.05 M K_2CO_3 concentration in the CO_2 liberation solution, K_2CO_3 (0.28 ± 0.005 g) was added at the end of each round, protected by the N_2 gas (Supplementary Fig. 76). This quantity was determined on the basis of stoichiometric calculations and quantified using a precise mass balance. Owing to a gradual decline in CE over the nine-round operation, the endpoint of each round was defined by the released CO_2 flow rate dropping below 0.2 sccm, indicating the near-complete conversion of CO_3^{2-} to gaseous CO_2. In addition, a concurrent decrease in the P1 and P2 voltages served as an auxiliary indicator of the reaction endpoint. Near the end of the CO_2 liberation process, the P1 voltage decreased due to the H^+ serving as the charge carrier, which possesses a low ionic resistance. Simultaneously, the P2 voltage dropped as CO_2 bubble formation at the P2 cathode ceased, thereby reducing the gas–liquid interfacial resistance. The duration of each round ranged from 7,700 to 8,500 s, reflecting the decrease in CE as the system approached higher OH^- concentrations.

A pre-oxidation step was implemented at the beginning of each day to restore the desired TEMPO-S:TEMPO-S$^+$ ratio that was required for both oxidation and reduction reactions. This step involved operating only P1 to allow P2 to reach 0.75 V at a current density of 50 mA cm^{-2}. In addition, we observed a net transfer of water (1.2 ± 0.1 ml) across the P1 CEM each day during electrochemical operation. This phenomenon is attributed to the migration of hydrated K^+ ions, which carry water molecules from the P1 anode compartment to the P1 cathode compartment. To compensate for this loss, DI water (1.2 ml) was added to the CO_2 liberation solution at the beginning of each day. Five electrochemically generated OH^- capture solutions were collected for UV-vis analysis to quantify the crossover of TEMPO-S through the CEM (Supplementary Fig. 78). In addition, the CO_2 liberation solution containing TEMPO-S was sampled every 20 h for UV-vis analysis to estimate its degradation rate (Supplementary Fig. 80).

Product analysis

To establish an equilibrium state in the fluid cycle, a 5-min purge of CO_2 gas into the CO_2 liberation solution was carried out before the electrochemical experiment, ensuring saturation of the CO_2 liberation solution with dissolved CO_2. The gas released from the gas–liquid separator was measured using a flow-meter (Omega FMA-LP1615A) with a customized vacuum pump connected to the outlet stream. During operation, the pH values of the capture solution and the CO_2 liberation solution were measured using the pH meters. Both solution temperatures were monitored during the experiment, maintaining a constant of 25 ± 0.5 °C.

The regenerated capture solution and CO_2 liberation solution were quantified using UV-vis spectrophotometry (PerkinElmer FL6500). The CO_2 liberation solution was diluted 100-fold before conducting the UV-vis analysis, whereas the UV-vis analysis of the regenerated capture solution was conducted without dilution. The TEMPO chemicals were examined using both high-resolution mass spectrometry and NMR spectroscopy. The ^{1}H NMR spectra of aqueous samples were collected using an internal standard, dimethyl sulfoxide (DMSO), with an Agilent DD2 500 spectrometer and water suppression mode.

During the TEMPO permeability tests in the H-cell configuration, samples in the receiving compartment were diluted 100-fold and then analyzed via UV-vis spectrophotometry. The spectral range was set from 200 to 400 nm with a scan rate of 240 nm min^{-1}.

A customized crystallizer was used for CO_2 capture experiments under atmospheric conditions (Supplementary Fig. 73). The weight of the rope before the conducting experiment and weight of the rope after the conducting experiment were measured using a high-resolution mass balance. The weight difference corresponded with the weight of salt precipitation.

The X-ray diffraction patterns for all samples were analyzed using a Rigaku MiniFlex diffractometer. The scan rate was set to 3° min^{-1}. The range of 2θ was set from 10 to 80°.

EPR analysis for all samples was carried out using a Bruker CW X-band ECS-EMXplus EPR Spectrometer. The resonator for routine EPR is an ER 4119HS, the liquid helium temperature control was set to 4 K and the liquid nitrogen temperature control was set in the range of 80–500 K.

Characterization via X-ray photoelectron spectroscopy was used to compare the elemental composition and oxidation states of the metal centers in the pristine electrodes and after the 200-hour stability tests.

Data availability

All data are available within the Article and its Supplementary Information. Source data are provided with this paper, including the atomic coordinates of the optimized computational models.

References

1. Bistline, J. E. T. & Blanford, G. J. Impact of carbon dioxide removal technologies on deep decarbonization of the electric power sector. *Nat. Commun.* **12**, 3732 (2021).
2. Erans, M. et al. Direct air capture: process technology, techno-economic and socio-political challenges. *Energy Environ. Sci.* **15**, 1360–1405 (2022).
3. Hashimoto, K. *Global Carbon Dioxide Recycling: For Global Sustainable Development by Renewable Energy* (Springer, 2019).
4. An, K., Li, K., Yang, C.-M., Brechtl, J. & Nawaz, K. A comprehensive review on regeneration strategies for direct air capture. *J. CO2 Util.* **76**, 102587 (2023).
5. Sanz-Pérez, E. S., Murdock, C. R., Didas, S. A. & Jones, C. W. Direct capture of CO_2 from ambient air. *Chem. Rev.* **116**, 11840–11876 (2016).
6. Deutz, S. & Bardow, A. Life-cycle assessment of an industrial direct air capture process based on temperature–vacuum swing adsorption. *Nat. Energy* **6**, 203–213 (2021).
7. Keith, D. W., Holmes, G., St. Angelo, D. & Heidel, K. A process for capturing CO_2 from the atmosphere. *Joule* **2**, 1573–1594 (2018).
8. Liu, A., Musgrave, C. B. III, Li, X., Goddard, W. A. III & Liu, Y. Non-aqueous alkoxide-mediated electrochemical carbon capture. *Nat. Energy* https://doi.org/10.1038/s41560-024-01614-7 (2024).
9. Xu, Y. et al. Regeneration of direct air CO_2 capture liquid via alternating electrocatalysis. *Joule* **7**, 2107–2117 (2023).
10. Eisaman, M. D. et al. CO_2 separation using bipolar membrane electrodialysis. *Energy Environ. Sci.* **4**, 1319–1328 (2011).
11. Yan, L., Bao, J., Shao, Y. & Wang, W. An electrochemical hydrogen-looping system for low-cost CO_2 capture from seawater. *ACS Energy Lett.* **7**, 1947–1952 (2022).
12. Liu, S. et al. Direct air capture of CO_2 via cyclic viologen electrocatalysis. *Energy Environ. Sci.* **17**, 1266–1278 (2024).
13. Zhu, P. et al. Continuous carbon capture in an electrochemical solid-electrolyte reactor. *Nature* **618**, 959–966 (2023).
14. Shu, Q., Legrand, L., Kuntke, P., Tedesco, M. & Hamelers, H. V. M. Electrochemical regeneration of spent alkaline absorbent from direct air capture. *Environ. Sci. Technol.* **54**, 8990–8998 (2020).
15. Zhang, X., Fang, Z., Zhu, P., Xia, Y. & Wang, H. Electrochemical regeneration of high-purity CO_2 from (bi)carbonates in a porous solid electrolyte reactor for efficient carbon capture. *Nat. Energy* **10**, 55–65 (2025).
16. Eisaman, M. D., Alvarado, L., Larner, D., Wang, P. & Littau, K. A. CO_2 desorption using high-pressure bipolar membrane electrodialysis. *Energy Environ. Sci.* **4**, 4031–4037 (2011).
17. Heidel, K., Keith, D., Singh, A. & Holmes, G. Process design and costing of an air-contactor for air-capture. *Energy Procedia* **4**, 2861–2868 (2011).
18. Salmón, I. R., Cambier, N. & Luis, P. CO_2 capture by alkaline solution for carbonate production: a comparison between a packed column and a membrane contactor. *Appl. Sci. (Basel)* **8**, 996 (2018).
19. Holmes, G. & Keith, D. W. An air–liquid contactor for large-scale capture of CO_2 from air. *Philos. Trans. R. Soc. Lond. A* **370**, 4380–4403 (2012).
20. Davis, S. M., Gray, G. E. & Kohl, P. A. Candidate membranes for the electrochemical salt-splitting of sodium sulfate. *J. Appl. Electrochem.* **38**, 777–783 (2008).
21. Liu, L., Li, H. & Avgouropoulos, G. A review of porous polytetrafluoroethylene reinforced sulfonic acid-based proton exchange membranes for fuel cells. *Int. J. Hydrogen Energy* **50**, 501–527 (2024).
22. Yang, G. et al. Organic electroactive materials for aqueous redox flow batteries. *Adv. Mater.* **35**, 2301898 (2023).
23. Luo, J., Hu, B., Hu, M., Zhao, Y. & Liu, T. L. Status and prospects of organic redox flow batteries toward sustainable energy storage. *ACS Energy Lett.* **4**, 2220–2240 (2019).
24. Zhang, L., Feng, R., Wang, W. & Yu, G. Emerging chemistries and molecular designs for flow batteries. *Nat. Rev. Chem.* **6**, 524–543 (2022).
25. Zhou, W. et al. Fundamental properties of TEMPO-based catholytes for aqueous redox flow batteries: effects of substituent groups and electrolytes on electrochemical properties, solubilities and battery performance. *RSC Adv.* **10**, 21839–21844 (2020).
26. Tang, B., Zhao, J., Xu, J.-F. & Zhang, X. Tuning the stability of organic radicals: from covalent approaches to non-covalent approaches. *Chem. Sci.* **11**, 1192–1204 (2020).
27. Feng, R., Chen, Y., Yang, L., Du, Q. & Zhuo, K. Ethyl viologen-functionalized reduced graphene oxide composites for asymmetric ionic liquid-based supercapacitors. *Chem. Eng. J.* **468**, 143693 (2023).
28. Yu, H.-F., Chen, K.-I., Yeh, M.-H. & Ho, K.-C. Effect of trifluoromethyl substituents in benzyl-based viologen on the electrochromic performance: optical contrast and stability. *Sol. Energy Mater. Sol. Cells* **200**, 110020 (2019).
29. Hu, L., Zhai, T., Li, H. & Wang, Y. Redox-mediator-enhanced electrochemical capacitors: recent advances and future perspectives. *ChemSusChem* **12**, 1118–1132 (2019).

30. Lv, X.-L. et al. Modular dimerization of organic radicals for stable and dense flow battery catholyte. *Nat. Energy* **8**, 1109–1118 (2023).
31. Fang, C. et al. Immobilizing an organic electrode material through π–π interaction for high-performance Li-organic batteries. *J. Mater. Chem. A* **7**, 22398–22404 (2019).
32. Chen, T., Li, M. & Liu, J. π–π stacking interaction: a nondestructive and facile means in material engineering for bioapplications. *Cryst. Growth Des.* **18**, 2765–2783 (2018).
33. Karlsson, C., Suga, T. & Nishide, H. Quantifying TEMPO redox polymer charge transport toward the organic radical battery. *ACS Appl. Mater. Interfaces* **9**, 10692–10698 (2017).
34. Zhou, Z. & Liu, L. TEMPO and its derivatives: synthesis and applications. *Curr. Org. Chem.* **18**, 459–474 (2014).
35. Winsberg, J. et al. Aqueous 2,2,6,6-tetramethylpiperidine-*N*-oxyl catholytes for a high-capacity and high current density oxygen-insensitive hybrid-flow battery. *ACS Energy Lett.* **2**, 411–416 (2017).
36. Gao, M., Salla, M., Song, Y. & Wang, Q. High-power near-neutral aqueous all organic redox flow battery enabled with a pair of anionic redox species. *Angew. Chem.* **134**, e202208223 (2022).
37. Beejapur, H. A. et al. TEMPO in chemical transformations: from homogeneous to heterogeneous. *ACS Catal.* **9**, 2777–2830 (2019).
38. Tai, R., Ogura, I., Okazaki, T., Iizumi, Y. & Mano, H. Acute toxicity tests of TEMPO-oxidized cellulose nanofiber using *Daphnia magna* and *Oryzias latipes*. *Cellulose* **31**, 2207–2220 (2024).
39. Pan, M. et al. Reversible redox chemistry in pyrrolidinium-based TEMPO radical and extended viologen for high-voltage and long-life aqueous redox flow batteries. *Adv. Energy Mater.* **12**, 2103478 (2022).
40. Ko, Y. et al. Redox mediators for oxygen reduction reactions in lithium–oxygen batteries: governing kinetics and its implications. *Energy Environ. Sci.* **16**, 5525–5533 (2023).
41. Mugisa, J., Chukwu, R., Brogioli, D. & La Mantia, F. Effect of ion-paring on the kinetics of redox systems with concentrated supporting electrolyte. *Electrochim. Acta* **473**, 143473 (2024).
42. Tang, G. et al. Adjusting Hirshfeld charge of TEMPO catholytes for stable all-organic aqueous redox flow batteries. *Nat. Commun.* **16**, 47 (2025).
43. Zhang, K., Lin, X., Shi, Y., Oyaizu, K. & Jia, Z. Two-electron redox chemistry of nitroxide radicals: fundamental mechanisms and applications in energy storage. *ACS Electrochem.* **1**, 123–137 (2025).
44. Zhang, Z. J. & Chen, X. Y. Illustrating the effect of electron withdrawing and electron donating groups adherent to *p*-hydroquinone on supercapacitor performance: the cases of sulfonic acid and methoxyl groups. *Electrochim. Acta* **282**, 563–574 (2018).
45. Xu, T. et al. Annular quaternary ammonium-grafted TEMPO for high-capacity and high-voltage aqueous redox flow batteries. *J. Power Sources* **635**, 236529 (2025).
46. Liu, T., Wei, X., Nie, Z., Sprenkle, V. & Wang, W. A total organic aqueous redox flow battery employing a low cost and sustainable methyl viologen anolyte and 4-HO-TEMPO catholyte. *Adv. Energy Mater.* **6**, 1501449 (2016).
47. Mukhopadhyay, A. et al. Mass transfer and reaction kinetic enhanced electrode for high-performance aqueous flow batteries. *Adv. Funct. Mater.* **29**, 1903192 (2019).
48. Zhong, H. et al. Air-stable and high-mobility n-channel organic transistors based on small-molecule/polymer semiconducting blends. *Adv. Mater.* **24**, 3205–3211 (2012).
49. Pedraza, E. et al. Unprecedented aqueous solubility of TEMPO and its application as high capacity catholyte for aqueous organic redox flow batteries. *Adv. Energy Mater.* **13**, 2301929 (2023).
50. Li, M. et al. Ionic liquid stabilized 2,2,6,6-tetramethylpiperidine 1-oxyl catalysis for alcohol oxidation. *ACS Sustain. Chem. Eng.* **8**, 4489–4498 (2020).
51. Prakash, N. et al. 2,2,6,6-Tetramethylpiperidinyloxyl (TEMPO) radical mediated electro-oxidation reactions: a review. *ChemistrySelect* https://doi.org/10.1002/slct.202102346 (2021).
52. Rabeah, J., Bentrup, U., Stößer, R. & Brückner, A. Selective alcohol oxidation by a copper TEMPO catalyst: mechanistic insights by simultaneously coupled operando EPR/UV-Vis/ATR-IR spectroscopy. *Angew. Chem. Int. Ed.* **54**, 11791–11794 (2015).
53. Huang, Z. et al. An "interaction-mediating" strategy towards enhanced solubility and redox properties of organics for aqueous flow batteries. *Nano Energy* **69**, 104464 (2020).
54. Orita, A., Verde, M. G., Sakai, M. & Meng, Y. S. The impact of pH on side reactions for aqueous redox flow batteries based on nitroxyl radical compounds. *J. Power Sources* **321**, 126–134 (2016).
55. Ma, Y., Loyns, C., Price, P. & Chechik, V. Thermal decay of TEMPO in acidic media via an *N*-oxoammonium salt intermediate. *Org. Biomol. Chem.* **9**, 5573–5578 (2011).
56. Sen, V. D. & Golubev, V. A. Kinetics and mechanism for acid-catalyzed disproportionation of 2,2,6,6-tetramethylpiperidine-1-oxyl. *J. Phys. Org. Chem.* **22**, 138–143 (2009).
57. Ding, C. et al. Morphology and electrochemical properties of perfluorosulfonic acid ionomers for vanadium flow battery applications: effect of side-chain length. *ChemSusChem* **6**, 1262–1269 (2013).
58. Qian, P., Wang, H., Jiang, Y., Zhou, Y. & Shi, H. High-performance composite membrane based on synergistic main-chain/side-chain proton conduction channels for the vanadium redox flow battery. *J. Mater. Chem. A* **9**, 4240–4252 (2021).
59. Tempelman, C. H. L., Jacobs, J. F., Balzer, R. M. & Degirmenci, V. Membranes for all vanadium redox flow batteries. *J. Energy Storage* **32**, 101754 (2020).
60. Grew, K. N. & Chiu, W. K. S. A dusty fluid model for predicting hydroxyl anion conductivity in alkaline anion exchange membranes. *J. Electrochem. Soc.* **157**, B327 (2010).
61. Zhao, Y. et al. Advanced charged sponge-like membrane with ultrahigh stability and selectivity for vanadium flow batteries. *Adv. Funct. Mater.* **26**, 210–218 (2016).
62. Clark, J. A., Santiso, E. E. & Frischknecht, A. L. Morphology and proton diffusion in a coarse-grained model of sulfonated poly(phenylenes). *J. Chem. Phys.* **151**, 104901 (2019).
63. Hickner, M. A., Ghassemi, H., Kim, Y. S., Einsla, B. R. & McGrath, J. E. Alternative polymer systems for proton exchange membranes (PEMs). *Chem. Rev.* **104**, 4587–4611 (2004).
64. Nagarale, R. K., Gohil, G. S. & Shahi, V. K. Sulfonated poly(ether ether ketone)/polyaniline composite proton-exchange membrane. *J. Membr. Sci.* **280**, 389–396 (2006).
65. Zhong, S., Fu, T., Dou, Z., Zhao, C. & Na, H. Preparation and evaluation of a proton exchange membrane based on crosslinkable sulfonated poly(ether ether ketone)s. *J. Power Sources* **162**, 51–57 (2006).
66. Mai, Z., Zhang, H., Li, X., Bi, C. & Dai, H. Sulfonated poly(tetramethydiphenyl ether ether ketone) membranes for vanadium redox flow battery application. *J. Power Sources* **196**, 482–487 (2011).
67. Liu, Y. et al. A long-lifetime all-organic aqueous flow battery utilizing TMAP-TEMPO radical. *Chem* **5**, 1861–1870 (2019).
68. Chen, X. et al. Spatially separated crystallization for selective lithium extraction from saline water. *Nat. Water* **1**, 808–817 (2023).
69. Nolte, O., Rohland, P., Ueberschaar, N., Hager, M. D. & Schubert, U. S. Stability of TMA-TEMPO-based aqueous electrolytes for redox-flow batteries. *J. Power Sources* **525**, 230996 (2022).
70. Schröter, E. et al. Oxidation of *N*,*N*,*N*,2,2,6,6-heptamethyl-piperidine-4-ammonium chloride to water-soluble *N*-oxyl radicals: a comparative study. *Eur. J. Org. Chem.* **2022**, e202200485 (2022).

71. Gerber, I. C. & Serp, P. A theory/experience description of support effects in carbon-supported catalysts. *Chem. Rev.* **120**, 1250–1349 (2020).
72. Shu, Q., Haug, M., Tedesco, M., Kuntke, P. & Hamelers, H. V. M. Direct air capture using electrochemically regenerated anion exchange resins. *Environ. Sci. Technol.* **56**, 11559–11566 (2022).
73. Li, X., Zhao, X., Liu, Y., Hatton, T. A. & Liu, Y. Redox-tunable Lewis bases for electrochemical carbon dioxide capture. *Nat. Energy* **7**, 1065–1075 (2022).
74. Wang, H. et al. A comprehensive and practical framework for advanced battery management system of vanadium redox flow batteries. *J. Energy Storage* **123**, 116560 (2025).
75. Alem, A. et al. Monitoring chemical processes in redox flow batteries employing in situ and in operando analyses. *Energy Environ. Sci.* **18**, 7373–7401 (2025).
76. Grignon, E., Hosseini, K., Berbigier, J. F., Edward, A. K. & Seferos, D. S. A low-permeability tempo-phosphate as an anionic posolyte for aqueous redox-flow batteries. *Angew. Chem. Int. Ed.* https://doi.org/10.1002/anie.202525352 (2026).

Acknowledgements

We thank M. J. Aziz and J. Parkey for discussion and D. Kopilovic for technical support, as well as J. Mudrik, R. Balazs and J. Shu for assistance in sample characterization. This work was financially supported by the Natural Resources Canada (NRCan) Energy Innovation Program, the Natural Sciences and Engineering Research Council of Canada (NSERC) via the Discovery Program, and TotalEnergies E&P Research & Technology USA, LLC. We acknowledge the support of the Government of Canada's New Frontiers in Research Fund (NFRF), CANSTOREnergy project, NFRFT-2022-00197, and the Canada Research Chairs Program (CRC-2021-00316). We also acknowledge many helpful inputs from members of the Sinton Group, as well as equipment and facilities throughout the labs, that made this work possible. We dedicate this paper to the memory of our co-author, Professor Dwight Seferos (1978–2026), whose discussion, mentorship and friendship will be greatly missed.

Author contributions

D. Sinton, Y.X., R.K.M. and D. Seferos supervised the project. S.L. and Y.X. conceived the electrochemical strategy. S.L. performed the electrochemical experiments. D.K. performed the crystallizing air capture unit experiments. E.G. performed the TEMPO synthesis. S.L. and E.G. carried out the TEMPO analysis and characterization. Z.G. and S.L. performed the membrane fabrication and analysis. Y.L. carried out the DFT simulations. Y.C.X., S.L. and Y.X. drafted the manuscript. Y.C.X., R.K.M., S.L., D.K., I.M., D. Sinton, J.P.E., P.P. and T.S. assisted with manuscript editing. J.L., Y.C. and F.L. discussed and assisted with the simulations. Y.C.X., D.K., H.Z., P.V.S., J. Zhu, R.K.M., J. Zhang, Q.W., Z.W., C.W. and H.S.L. contributed to the data and product analysis, and the experimental discussion. All authors discussed the results and assisted during manuscript preparation.

Competing interests

A US patent application (no. 10004575), entitled 'Cyclic organic redox-active mediators assisted direct air capture and release CO_2', is under disclosure by D. Sinton, Y.X., S.L. and D.K.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44286-026-00391-2.

Correspondence and requests for materials should be addressed to Rui Kai Miao, Yi Xu or David Sinton.

Peer review information *Nature Chemical Engineering* thanks Kiana Amini, Yayuan Liu and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.